

Statin therapy induces ultrastructural damage in skeletal muscle in patients without myalgia.

Muscle pain and weakness are frequent complaints in patients receiving 3-hydroxymethylglutaryl coenzymeA (HMG CoA) reductase inhibitors (statins). Many patients with myalgia have creatine kinase levels that are either normal or only marginally elevated, and no obvious structural defects have been reported in patients with myalgia only. To investigate further the mechanism that mediates statin-induced skeletal muscle damage, skeletal muscle biopsies from statin-treated and non-statin-treated patients were examined using both electron microscopy and biochemical approaches. The present paper reports clear evidence of skeletal muscle damage in statin-treated patients, despite their being asymptomatic. Though the degree of overall damage is slight, it has a characteristic pattern that includes breakdown of the T-tubular system and subsarcolemmal rupture. These characteristic structural abnormalities observed in the statin-treated patients were reproduced by extraction of cholesterol from skeletal muscle fibres in vitro. These findings support the hypothesis that statin-induced cholesterol lowering per se contributes to myocyte damage and suggest further that it is the specific lipid/protein organization of the skeletal muscle cell itself that renders it particularly vulnerable. Copyright (c) 2006 Pathological Society of Great Britain and Ireland. Published by John Wiley & Sons, Ltd.